We've gone through ways to save feature tables, how to build streaming features, implement point-in-time lookups, and how to create on-demand features. It's time to get ready to follow along in your own Databricks workspace as we work through the *Chapter 5* project code.

Applying our learning

Now that we have learned about the feature engineering components of the DI platform, let's put these topics into action and build on the example project datasets with new features that will enhance the data science projects.

Technical requirements

Here are the technical requirements needed to complete the hands-on examples in this chapter:

- The *Streaming Transactions* project requires more compute power than is available in the single node cluster. We created a multi-node cluster to address this. See *Figure 5.4* for the multi-node CPU configuration we used.

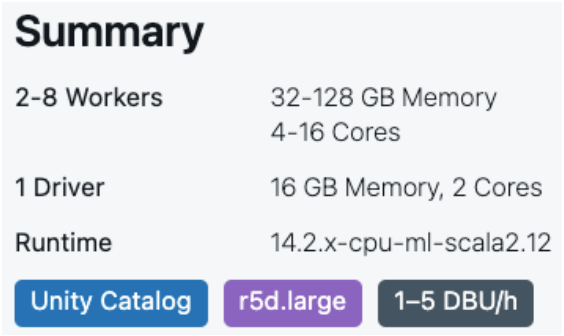


Figure 5.4 – Multi-node CPU cluster configuration (on AWS) used for this book

- We will use managed volumes to store cleaned, featurized data.

Project: Streaming Transactions

If you've been following the code in the previous chapters, at this point, you have the streaming data you need. In this chapter, we will augment that data with some of the feature engineering techniques discussed earlier. First, we will create a streaming feature to count the number of transactions that have arrived in the last two minutes for each customer.

Before we jump in, let's remember where we are and where we are going.